

Basic Principles of Medicine 2
Module: DM
Lecture No: 23

Ammonia Metabolism and Urea cycle

Dr. Sharmila Upadhya
shupadhya@sgu.edu
St Georges University ©

Copyright

All year 1 courses materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed or published in any form, printed, electronic or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

Required pre-reading and objectives

- Before DM Lecture 23

Objectives: DLA: Introduction to Nitrogen Metabolism and Fates of Amino Acids

SOM. MK.I.BPM2.3.DM.2.BCHM.1216	Analyze how amino acids may enter intermediary metabolism and describe 'amino acid pool'
SOM. MK.I.BPM2.3.DM.2.BCHM.1217	Discuss the reactions of alanine amino transferase, aspartate aminotransferase, glutaminase and glutamate dehydrogenase and analyze the central role of these amino acids in metabolism
SOM. MK.I.BPM2.3.DM.2.BCHM.1218	Define and enumerate the essential and nonessential amino acids in humans
SOM. MK.I.BPM2.3.DM.2.BCHM.1220	Distinguish the fates of C- skeletons of amino acids: Glucogenic amino acids, Ketogenic amino acids, Glucogenic & Ketogenic amino acids with examples.
SOM. MK.I.BPM2.3.DM.2.BCHM.1221	Review the TCA cycle intermediates formed from glutamate, glutamine, aspartate, and alanine.

Recommended Reading

- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- **Lippincott's Illustrated reviews: Biochemistry 9e**
 - **Amino acid degradation and synthesis:** [Amino Acids: Degradation and Synthesis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#) Questions: 20.1-20.7
 - **Disposal of Nitrogen and Urea Cycle:** [19. Amino Acids: Nitrogen Disposal | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#) Questions: 19.1-19.5
- **Concept map: Fig. 19.21**

Objectives: Ammonia metabolism and Urea cycle

SOM. MK.I.BPM2.3.DM.2.BCHM.1222	Explain the formation of urea in the liver and its excretion via the kidneys.
SOM. MK.I.BPM2.3.DM.2.BCHM.1223	Compare and contrast the inherited disorders associated with the urea cycle highlighting the enzyme deficient, laboratory findings, and clinical features of hyperammonemia.
SOM. MK.I.BPM2.3.DM.2.BCHM.1224	Identify the transport forms of ammonia from peripheral tissues (alanine & glutamine). Indicate the role of glutamate dehydrogenase and glutamine synthetase in glutamine formation.
SOM. MK.I.BPM2.3.DM.2.BCHM.1225	Describe the transamination (aminotransferase) reaction and the role of vitamin B6 (pyridoxal phosphate).
SOM. MK.I.BPM2.3.DM.2.BCHM.1226	Describe the role of glutamate dehydrogenase in donating NH_3 for the urea cycle. Identify the sources of nitrogen for the formation of urea
SOM. MK.I.BPM2.3.DM.2.BCHM.1227	List the reactions that form ammonia in the liver: Explain formation of ammonia in the intestine and its detoxification. Analyze the significance of intestinal ammonia formation in patients with compromised liver function.
SOM. MK.I.BPM2.3.DM.2.BCHM.1228	Indicate the metabolic basis for the use of phenylacetate, benzoic acid, arginine and carbamoyl glutamate (N-acetyl glutamate) in the management of the urea cycle disorders
SOM. MK.I.BPM2.3.DM.2.BCHM.1229	Analyze the mechanisms of neurotoxicity of hyperammonemia. Explain the metabolic basis of use of antibiotics and lactulose in the management of acquired hyperammonemia

Overview of amino acid metabolism

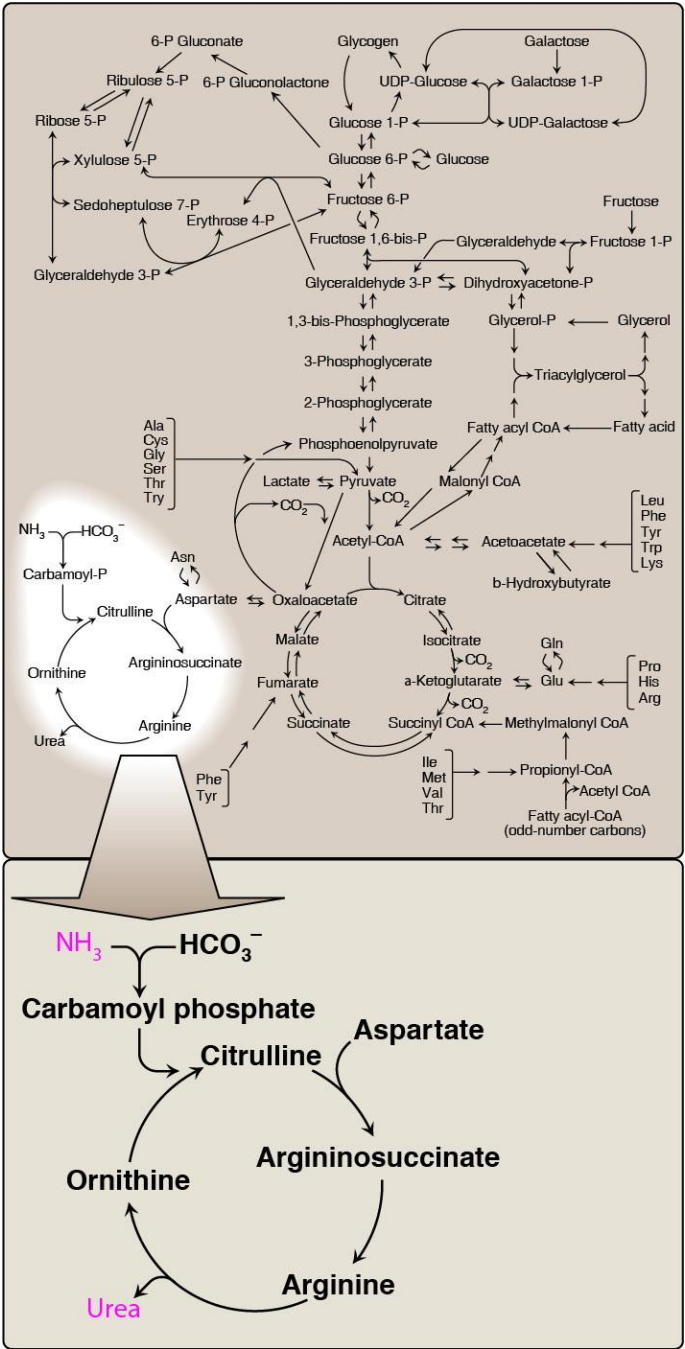
Amino Acid metabolism

Catabolism of amino acids

Carbon skeleton:
Glucogenic
or Ketogenic
amino acids

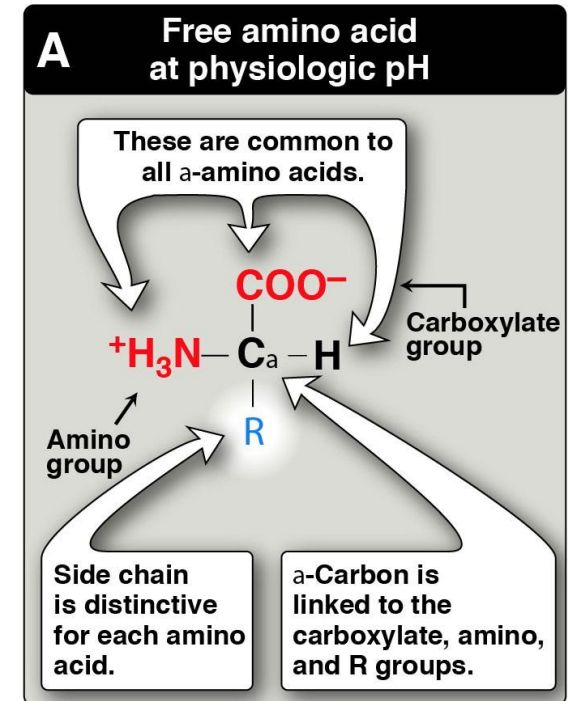
Ammonia formation
& Urea cycle

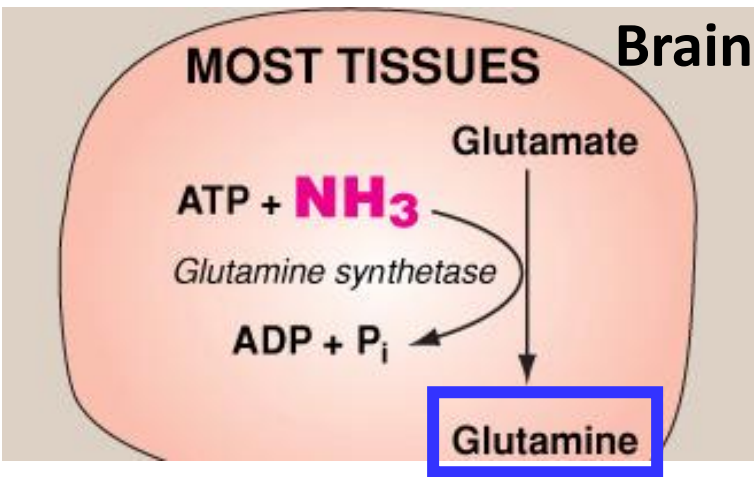
Specialized products
of amino acids
(Heme, purine, pyrimidines,
creatine)



Transport of Nitrogen from peripheral tissues

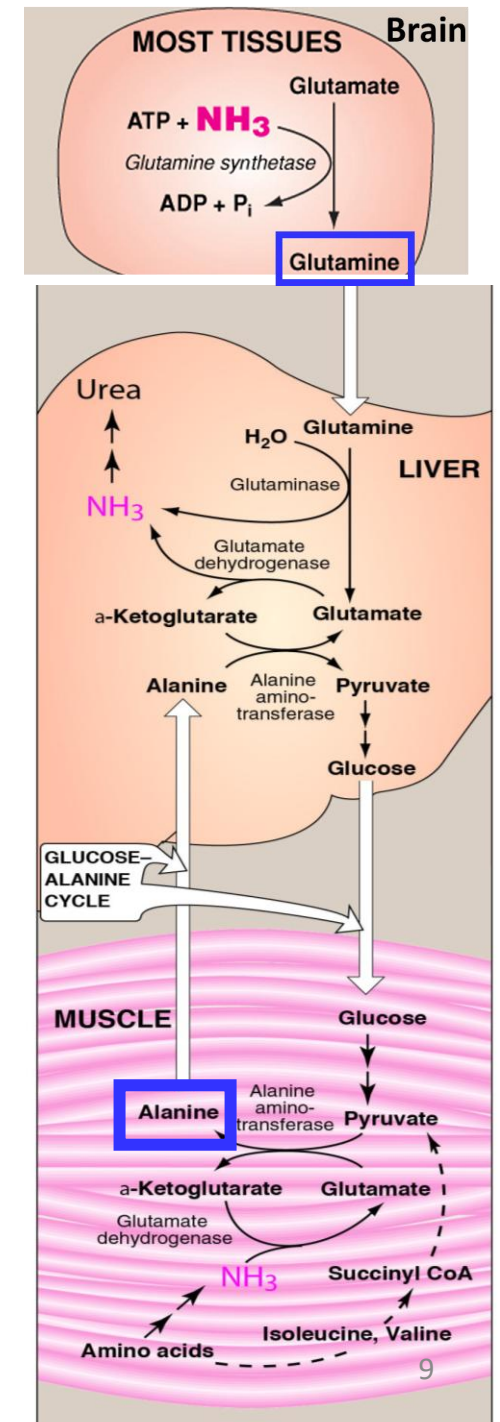
- Ammonia is neurotoxic
 - Creates requirement for a non-toxic transport form of ammonia
 - **In Urea cycle defects and severe liver disease: Blood ammonia elevated**
- **N transported from peripheral tissues in non-toxic form**
 - **Glutamine** (Incorporation of NH_3 to glutamate)
 - In Urea cycle defects: Glutamine elevated
 - **Alanine** (Transport form of NH_3 from muscle)





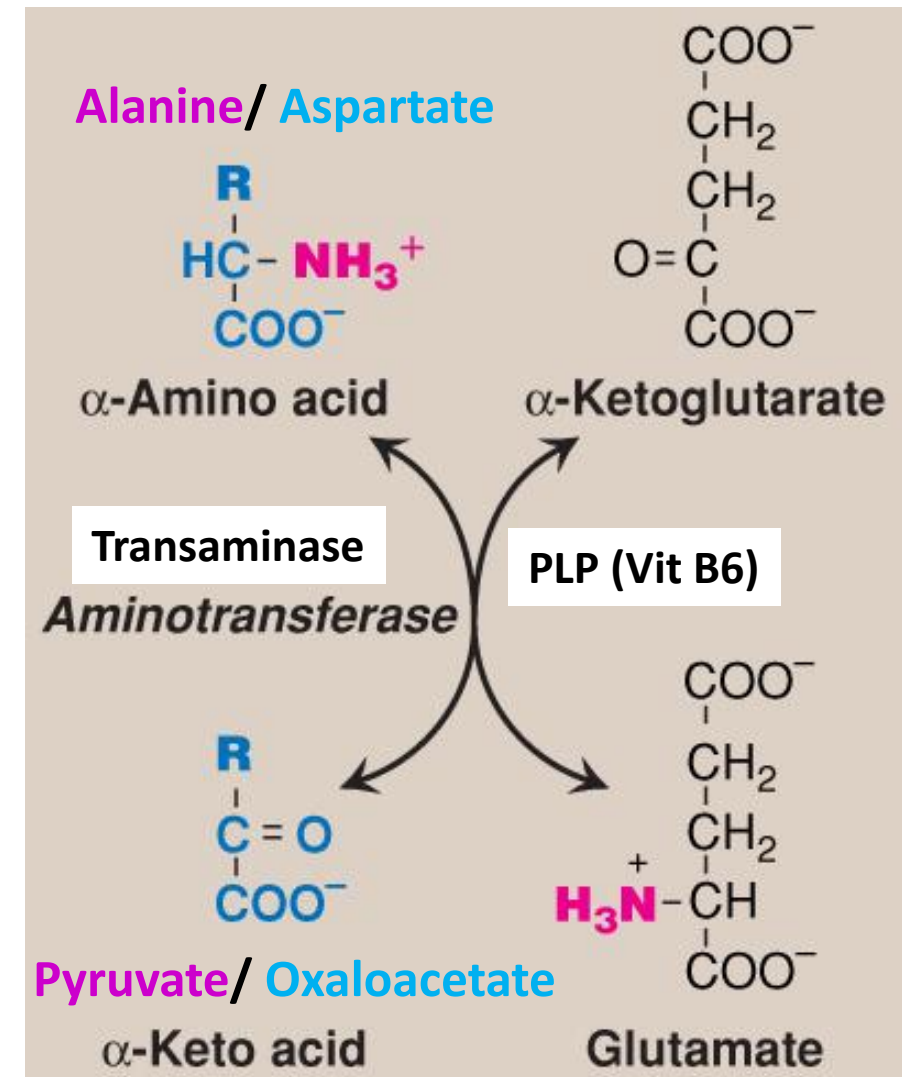
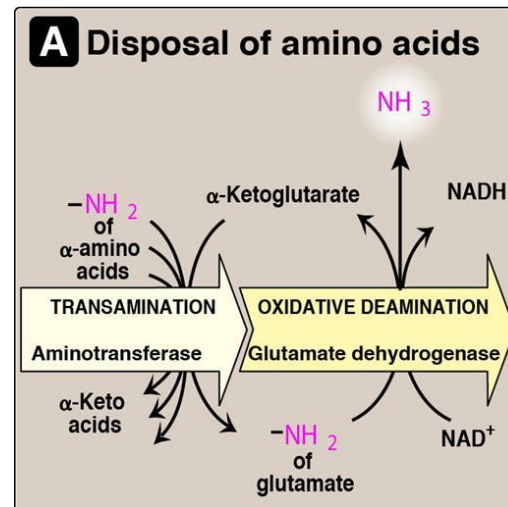
Ammonia transport

- Glutamine formed in most tissues (**Brain**):
 - Synthesized from glutamate
 - Enzyme: Glutamine synthetase
 - **In Urea cycle defects: Glutamine elevated**
- Alanine: Synthesized in **muscle**
 - Transamination from pyruvate
 - Enzyme: ALT



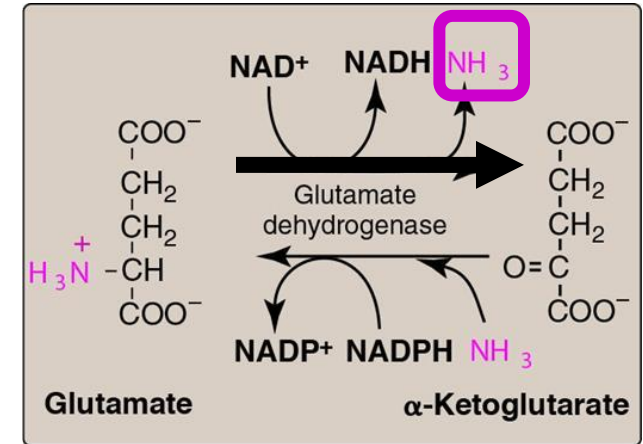
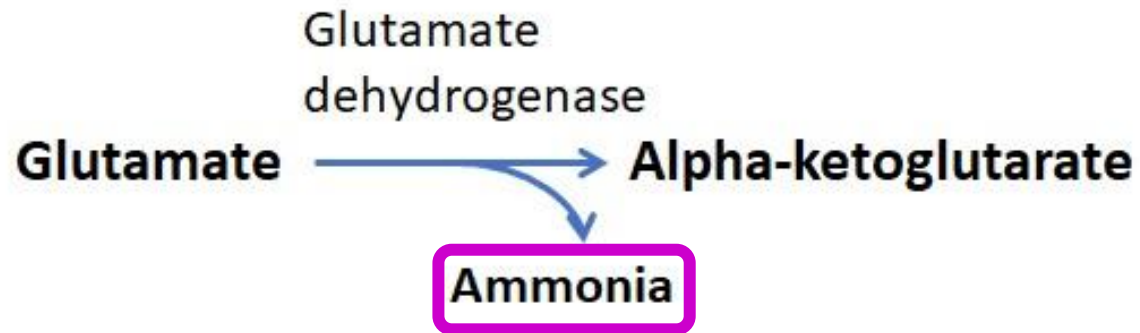
Aminotransferase reaction

- Most amino acids undergo transamination with α -KG as the amino acceptor
- **Amino group transfer** (Free ammonia **not released**)
- Two substrate used in the reaction (and 2 products formed): **Review DLA**
- **ALT and AST**
- Glutamate is formed
- Amino group transferred from amino acids to **Glutamate**
- Requires **PLP**
- **Alanine $\rightarrow$ Pyruvate**
- **Aspartate $\rightarrow$ Oxaloacetate**



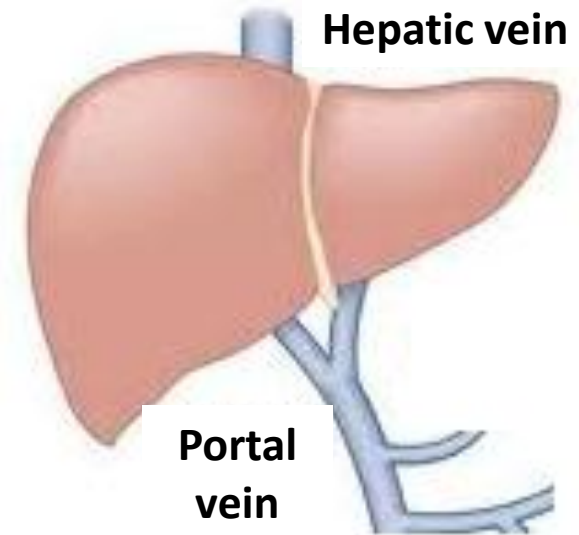
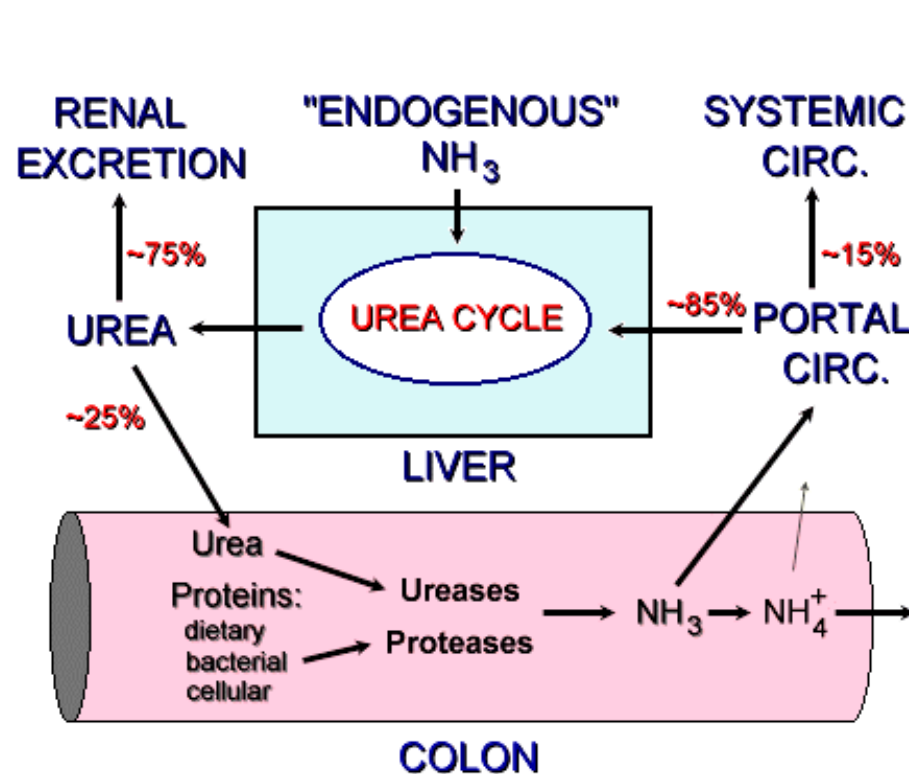
Ammonia (NH₃) formation in liver

- By **Glutamate dehydrogenase**
 - Forms **free ammonia** → **Enters Urea cycle**



- From Glutamine
 - By **Glutaminase** reaction, free ammonia and glutamate are formed
 - **Glutaminase in renal tubule** forms NH₃ → NH₄⁺ (Lost in urine): Important for Acid-base balance and H⁺ excretion

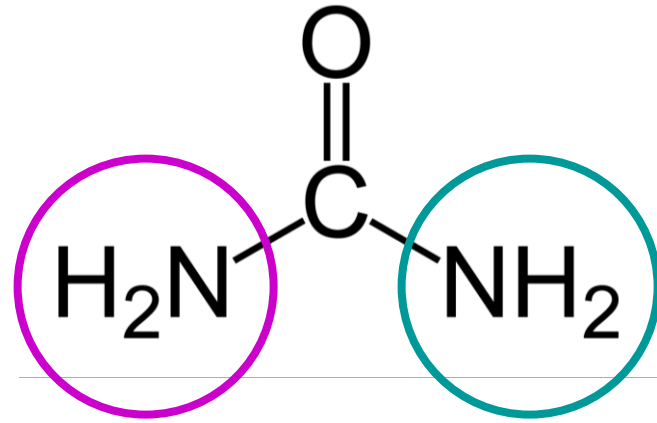
Ammonia (NH₃) formation in the gut and fate



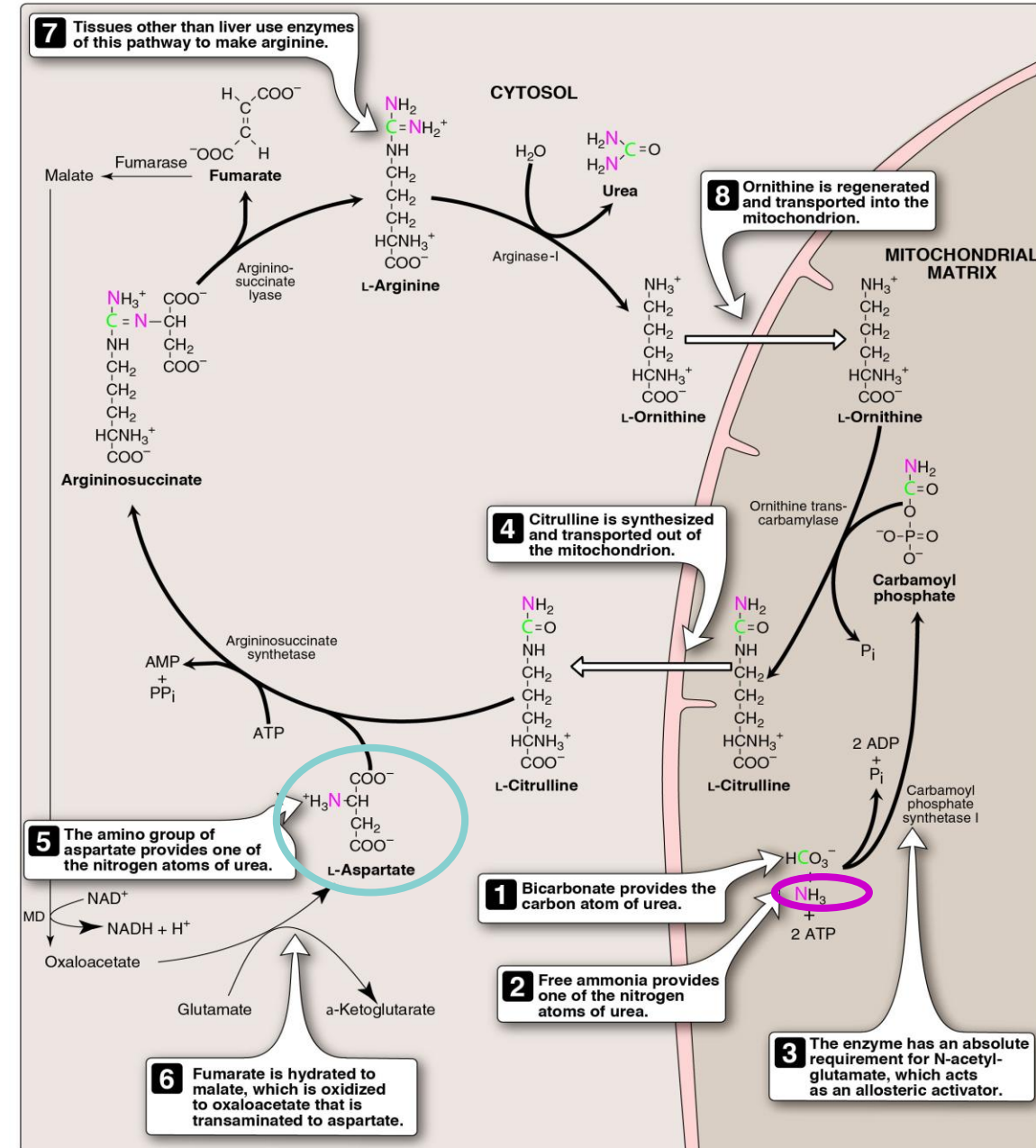
Cirrhosis increases the amount of ammonia entering systemic circulation

- Bacterial ureases form ammonia in colon.
- Ammonia enters portal circulation and forms **urea in liver**.
- Intestinal ammonia formation is significant in **liver disease** (ammonia not detoxified) → Ammonia enters systemic circulation resulting in neurotoxicity

Urea cycle

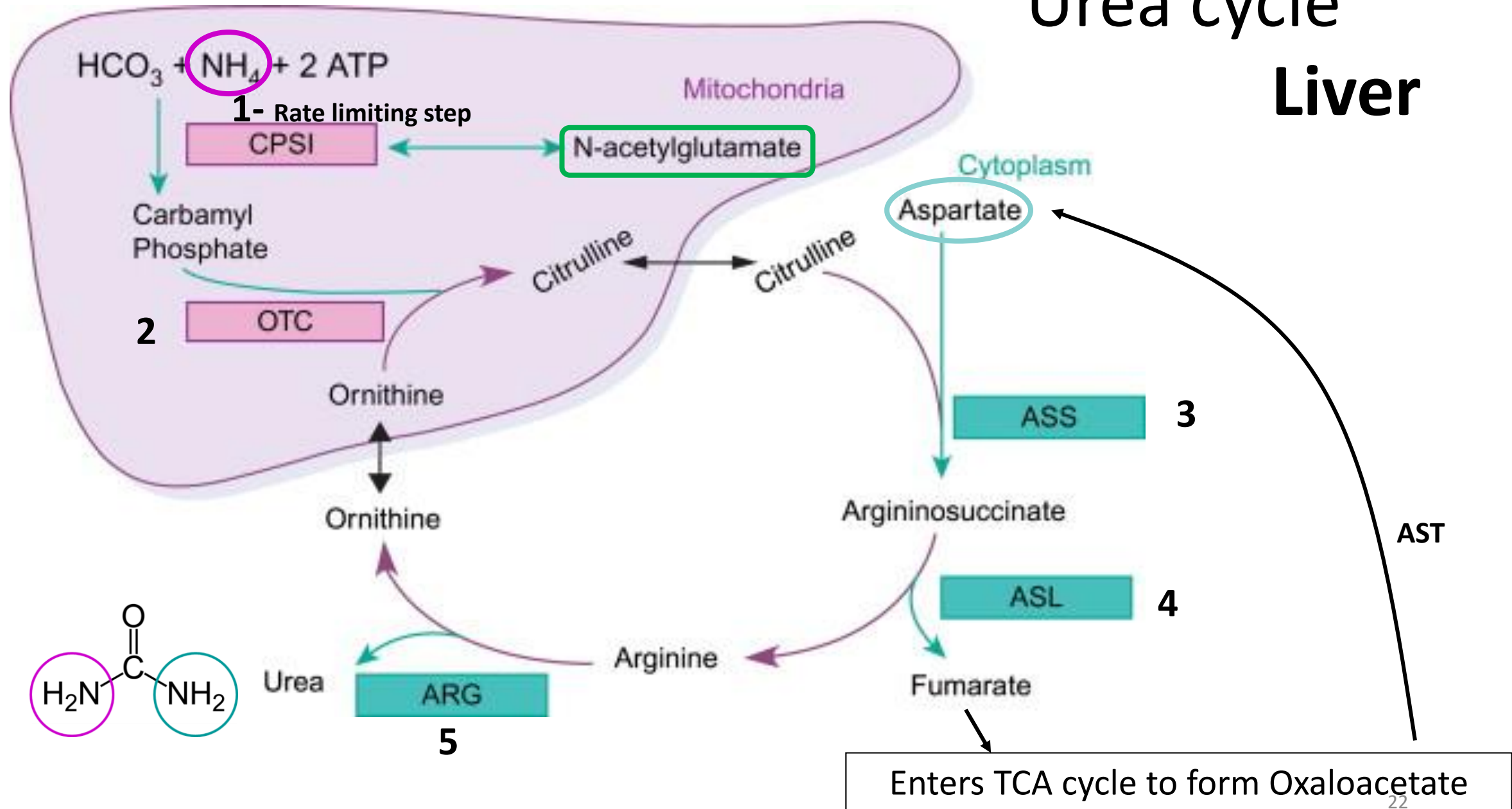


- Takes place in liver
- Partly mitochondrial (two reactions) and partly cytosolic (three reactions)
- Urea contains **two nitrogen atoms**
 - First N atom from **Ammonia**
 - Second N atom from **Aspartate**
- **Sources of nitrogen (Glutamate dehydrogenase)** and formation of aspartate from oxaloacetate



Urea cycle

Liver



Reactions of the urea cycle (Memorize pathway)

1 – CPS-I: Carbamoyl phosphate synthetase-I

- Mitochondrial
- Activated by **N-acetyl glutamate (NAG)**
- Incorporates free **ammonia**
- Do not confuse with **CPS-II (pyrimidine nucleotide synthesis)** and **CPT-I (beta-oxidation and carnitine shuttle)**

2 - OTC: Ornithine transcarbamoylase

- Mitochondrial

3 - ASS: Argininosuccinate synthetase

- Uses **Aspartate**

4 - ASL: Argininosuccinate lyase

5 - ARG: Arginase

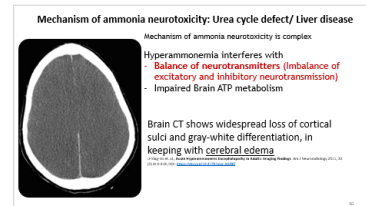
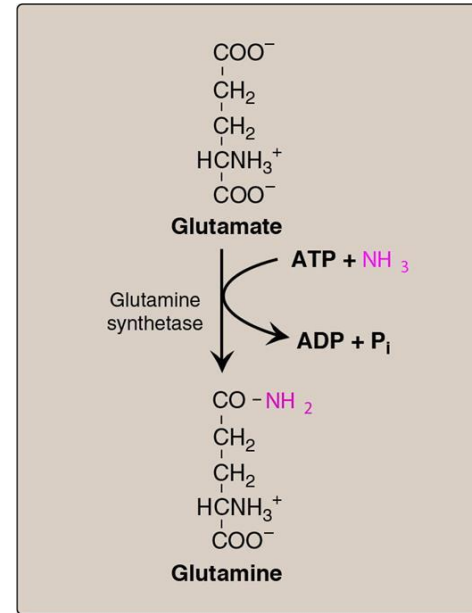
- **Forms Urea**

Hyperammonemia and Urea cycle defect

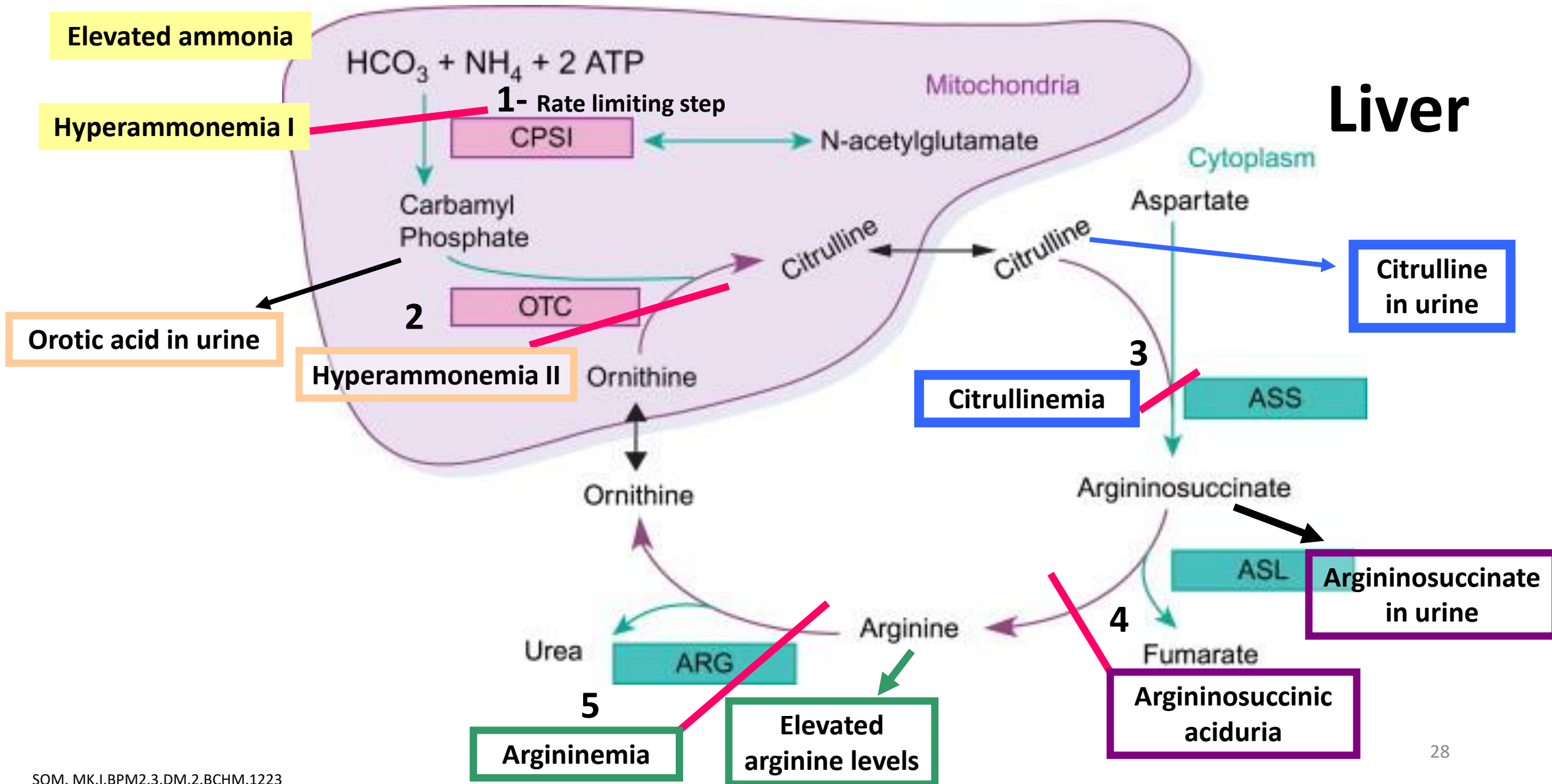
- A 36-hour-old male newborn starts vomiting, develops grunting respiration (hyperventilation) and rapidly became lethargic and unresponsive (neurological).
- Born at full-term after a normal pregnancy.
- Parents were first cousins, and it was the couple's first pregnancy.
- Metabolic screen: **Elevated serum ammonia and glutamine levels**
- Laboratory studies show **respiratory alkalosis**
 - pH- Increased; PCO₂- Decreased; HCO₃⁻ - Normal or low (Acute or chronic)
 - Ammonia directly stimulates respiratory center → Hyperventilation → Respiratory alkalosis
- Plasma urea at the lower end of normal

Inherited disorders of Urea cycle

- Enzyme deficiency → Accumulation of SUBSTRATE of enzyme
- **Increased blood ammonia** (hyperammonemia) and **glutamine** (Favors Glutamate → Glutamine)
- CPS-1 or OTC Deficiency (first two enzymes) most severe
- In all disorders, urea formation is decreased
- Hyperammonemia symptoms appear during newborn/infancy/childhood:
 - Lethargy, Irritability, Feeding difficulties (ammonia neurotoxicity)
 - Respiratory alkalosis (Ammonia stimulates respiratory center → hyperventilation)
- Neurological manifestations, seizures and intellectual disability, and developmental delay (due to elevated ammonia levels)



Urea cycle Disorders (UCD)

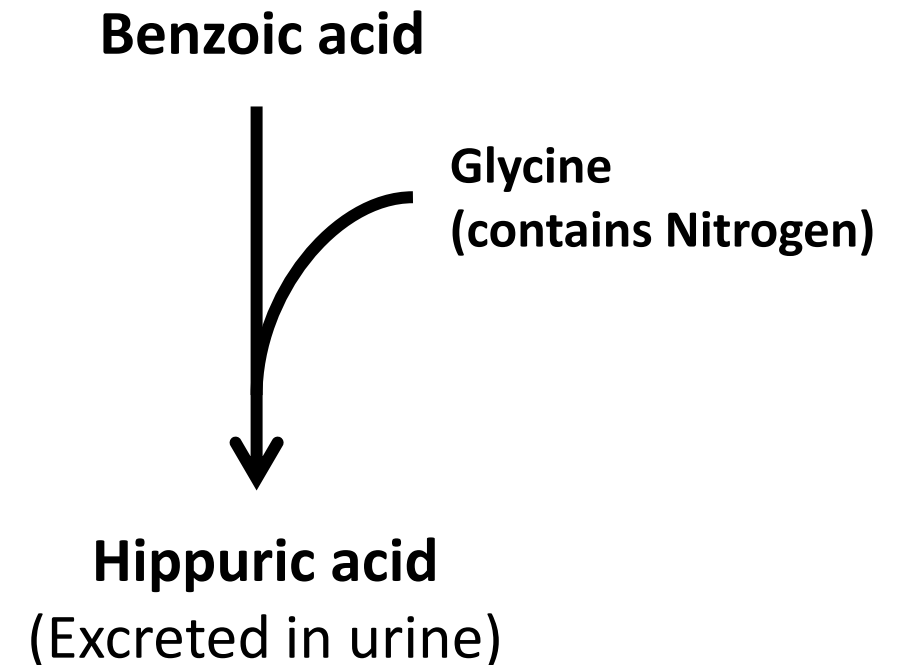
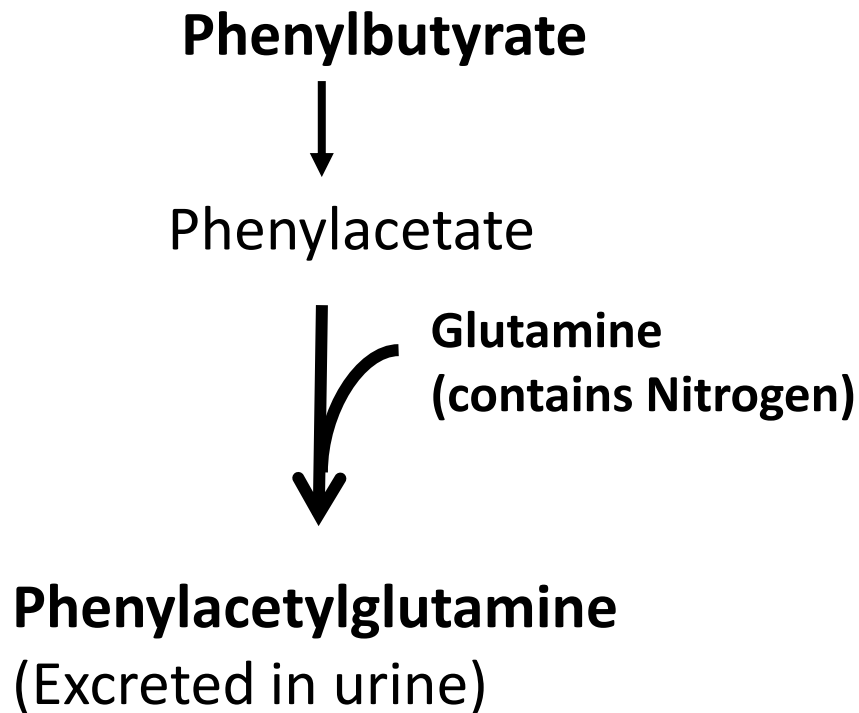


Management of hyperammonemia (**All UCD**)

- Dialysis: acute emergency
- Low protein / high carbohydrate
 - Minimize N intake
- Prevention of stress that induces a catabolic state
- Administration of **benzoic acid** and or **Phenylbutyrate / phenylacetate** (alternate routes of N excretion)
- Arginine administration
- Long term: Liver transplantation

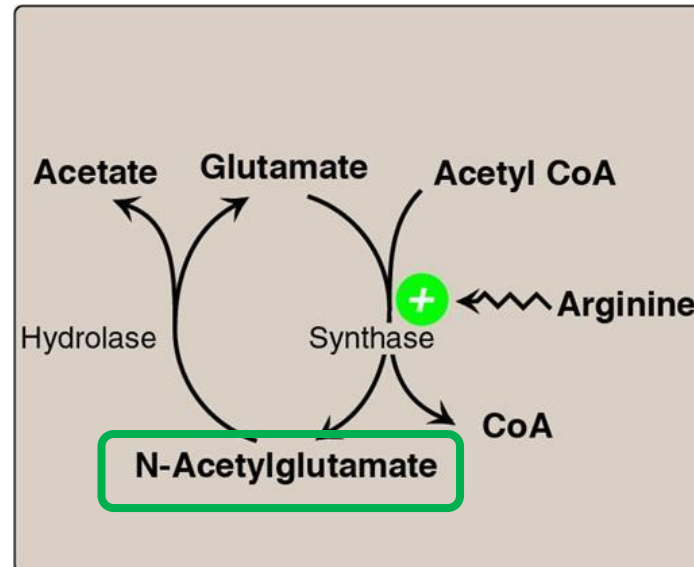
Management of hyperammonemia: Benzoic acid and Phenylbutyrate

- **Phenylbutyrate/Phenylacetate** + Glutamine (N) forms Phenylacetylglutamine
- **Benzoic acid** + Glycine (N) forms Hippuric acid
- Excreted in urine (Nitrogen loss)



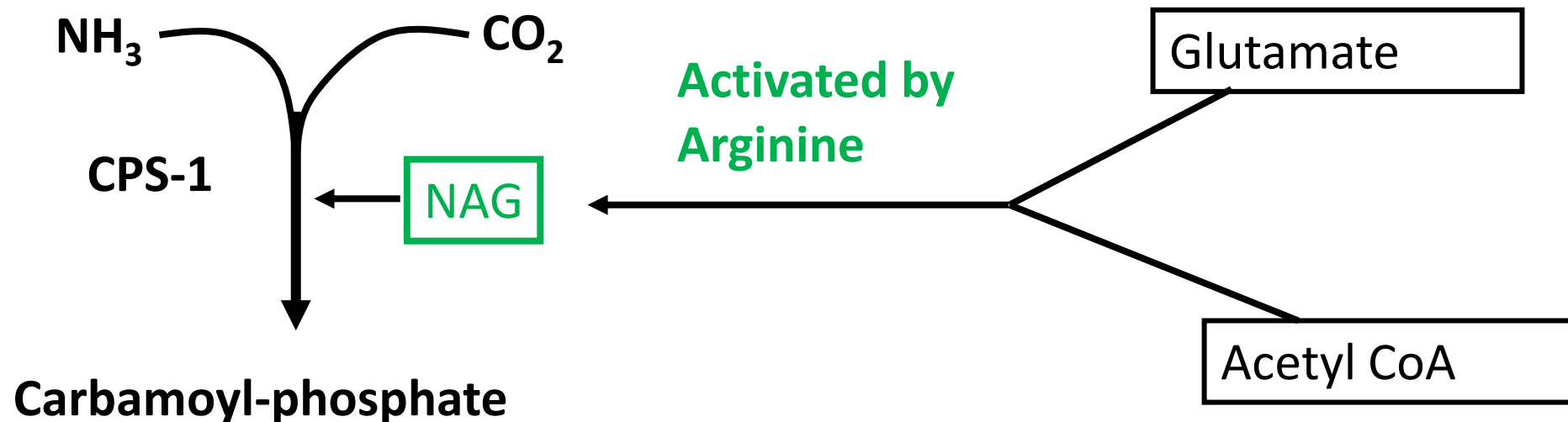
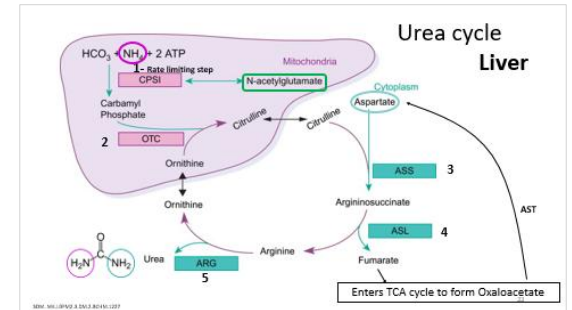
Management of hyperammonemia: Arginine administration

- For all the UCD (**except, for arginase deficiency**):
- Treatment *may* include **arginine** administration:
 - Allows urea cycle to progress by increasing substrate concentration
 - Arginine → Ornithine (arginase enzyme)
- **Arginine** activates **NAG** synthesis (NAG allosteric activator of **CPS-1**)
- NAG: **N-acetyl glutamate**



CPS-1 deficiency (Type I Hyperammonemia)

- **Very high blood ammonia and glutamine levels; Most severe**
- **Low levels of ALL urea cycle intermediates**
- **Sometimes** responds to Arginine supplements
 - Rationale: Arg stimulates formation of **N-acetylglutamate**
 - High levels of NAG *might* stimulate deficient CPS-I
 - Remember, NAG (Allosteric activator of CPS-I)



OTC deficiency (X-linked)

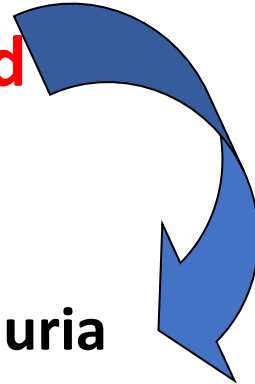
- Hyperammonemia Type II
 - **Most common UCD**
 - Usually seen in males, and more severe in males (**X-linked**)
 - Blood glutamine elevated
- **Mitochondrial ornithine transcarbamoylase**
- Normally:
 - OTC combines ornithine and carbamoyl phosphate to form citrulline
 - Citrulline is transported to cytoplasm
 - Citrulline combines with aspartate to form arginosuccinate

Ornithine transcarbamoylase deficiency - OTC

(Type-II Hyperammonemia)

- Elevated serum ammonia and glutamine
- **Elevated serum & *urine* Orotic acid**

Orotic aciduria



Elevated carbamoyl phosphate drives **Pyrimidine biosynthesis** →
Orotic acid elevated in serum & urine

Treatments listed previously – of course

Argininosuccinate synthetase deficiency (Citrullinemia)

- Diagnosis: Hyperammonemia, high glutamine levels; Very high levels of serum and urinary **citrulline**
- Treatments listed previously (of course)

Argininosuccinate lyase deficiency (Argininosuccinic aciduria)

- Differential diagnosis of hyperammonemia
 - Elevated ammonia and glutamine
 - **Argininosuccinate** elevated in plasma and urine
 - Also, elevated **citrulline** (because it is before the block)
- Treatments listed previously – of course

Arginase deficiency (Hyperargininemia)

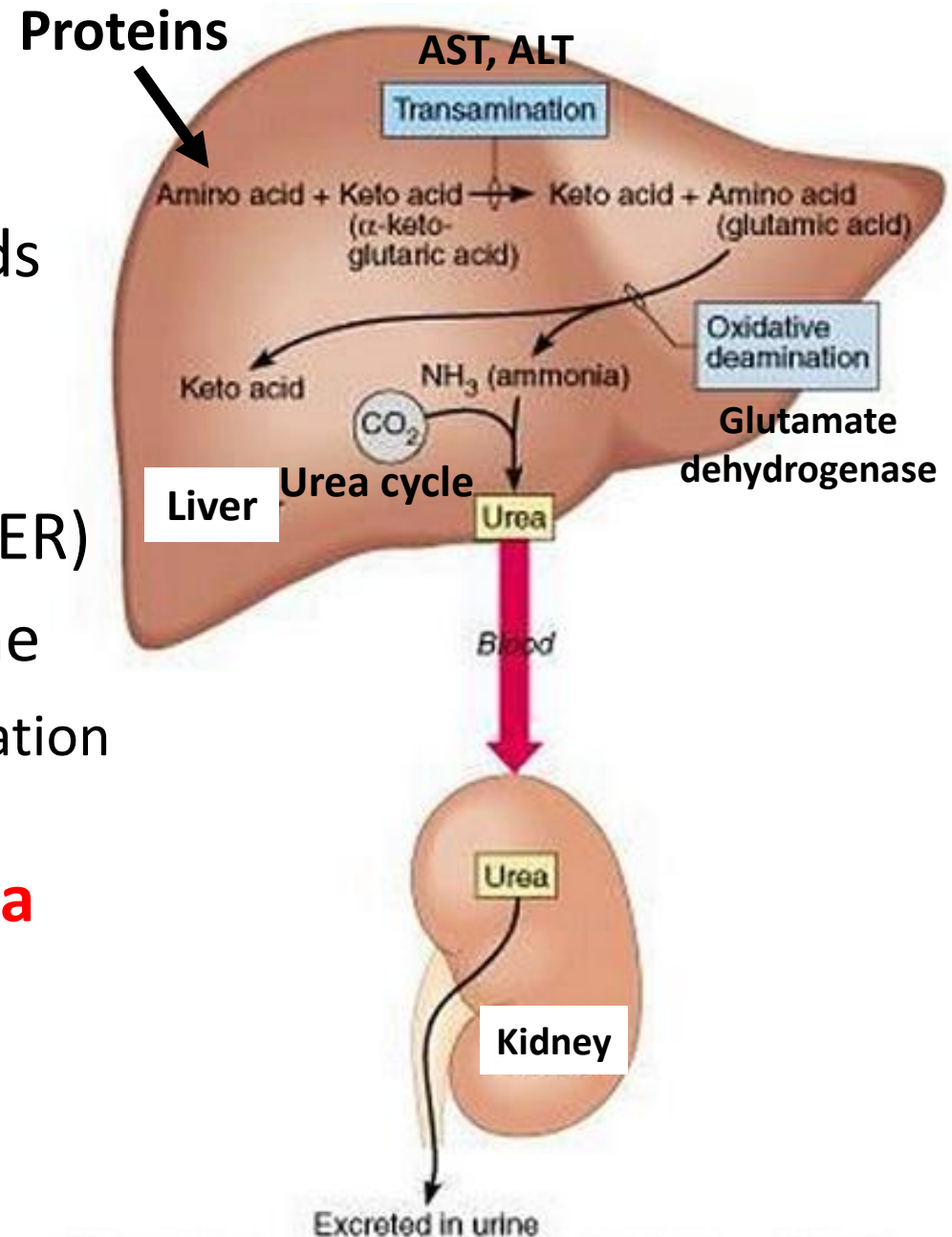
- Elevated serum ammonia and elevated arginine
 - Blood NH_3 levels not as high as other UCD
- Treatment includes diet of essential amino acids *excluding* arginine
- Other treatments previously listed (of course)
- Significant number of cases are adult onset
 - Associated with neurological problems

Inherited Urea Cycle Disorders (UCD)

Disease	Enzyme deficient	Lab findings	Management/ Treatment (explain basis)
Hyperammonemia Type I	CPS I	Hyperammonemia (most severe); Elevated glutamine; Low levels of all urea cycle intermediates	Low protein diet; Phenylacetate; Benzoic acid; Arginine
Hyperammonemia Type II	OTC (X-linked; Most common)	Hyperammonemia; Elevated glutamine; Increased orotic acid excretion in urine (Explain)	Low protein diet; Phenylacetate; Benzoic acid; Arginine
Citrullinemia	Argininosuccinate synthetase	Hyperammonemia; Elevated glutamine; Increased citrulline levels	Low protein diet; Phenylacetate; Benzoic acid; Arginine
Argininosuccinic aciduria	Argininosuccinate lyase (ASL)	Hyperammonemia; Elevated glutamine; Increased argininosuccinate levels; Mild elevation in citrulline	Low protein diet; Phenylacetate; Benzoic acid; Arginine
Argininemia	Arginase	Mild hyperammonemia; Elevated glutamine; Increased arginine levels	Low protein diet; Phenylacetate; Benzoic acid; Arginine

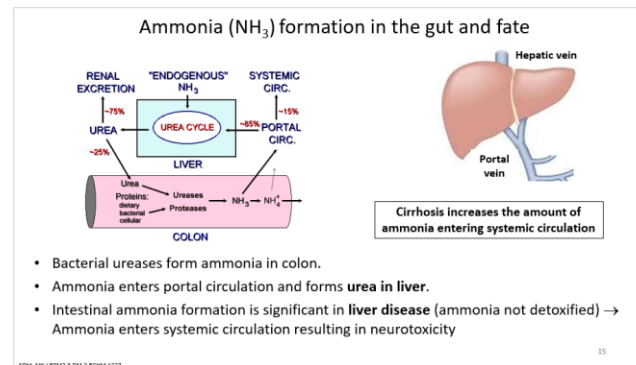
Formation and Fate of Urea

- Endogenous and dietary proteins → Amino acids
- Amino acids → Ammonia (transamination and oxidative deamination)
- Ammonia (derived from **proteins**) → UREA (LIVER)
- Urea transported to kidney and excreted in urine
- Some urea is degraded in gut, ammonia reenters circulation and the liver must detoxify it
- **Renal failure leads to elevated BUN (Blood urea nitrogen)**
- **Elevated blood ammonia**
 - Severe Liver disease
 - Inherited Urea cycle defects



Acquired hyperammonemia

- **Liver disease** due to viral or drug induced hepatitis, alcoholic cirrhosis
- In cirrhosis, there is porto-systemic shunting of blood
 - Portal blood enters the systemic circulation without going to the liver
 - Ammonia produced in intestine directly enters circulation and results in neurotoxicity



Acquired hyperammonemia scenario

- A 56-year-old man with a 1-month history of abdominal distension, **confusion, inability to concentrate.**
 - Diagnosed with alcohol-induced liver cirrhosis 4 yrs ago
 - History of excessive alcohol use for approx 20 yrs.
- On examination, he is **apathetic, confused, disoriented, drowsy, slurred speech. Flapping tremor (asterixis)** [Asterixis \(AKA Flapping Tremor\)](#)
- Cirrhosis from alcoholism, severe hepatitis
 - Liver not able to detoxify NH_3 in circulation
 - Elevated **Blood ammonia levels**
 - Elevated **plasma glutamine levels**

Treatment of acquired hyperammonemia: Basis

- Low-protein/high carb diet → intuitive
 - Less N ingested, less needed to dispose
- **Lactulose** → A disaccharide
 - Resistant to digestion in the small intestine
 - Normal flora digest lactulose, produce lactic acid, lactic acid is neutralized by NH_4^+
 - More N excreted in the feces
- Neomycin → (or other antibiotic treatment)
 - Reduction in bacterial urease in the gut
- Might use some of the other treatments previously discussed

Mechanism of ammonia neurotoxicity: Urea cycle defect/ Liver disease

Mechanism of ammonia neurotoxicity is complex

Hyperammonemia interferes with

- **Balance of neurotransmitters** (Imbalance of excitatory and inhibitory neurotransmission)
- Impaired Brain ATP metabolism



Brain CT shows widespread loss of cortical sulci and gray-white differentiation, in keeping with cerebral edema

U-King-Im et. al., **Acute Hyperammonemic Encephalopathy in Adults: Imaging Findings** Am J Neuroradiology 2011, 32 (2) 413-418; DOI: <https://doi.org/10.3174/ajnr.A2290>

Mechanism of ammonia neurotoxicity

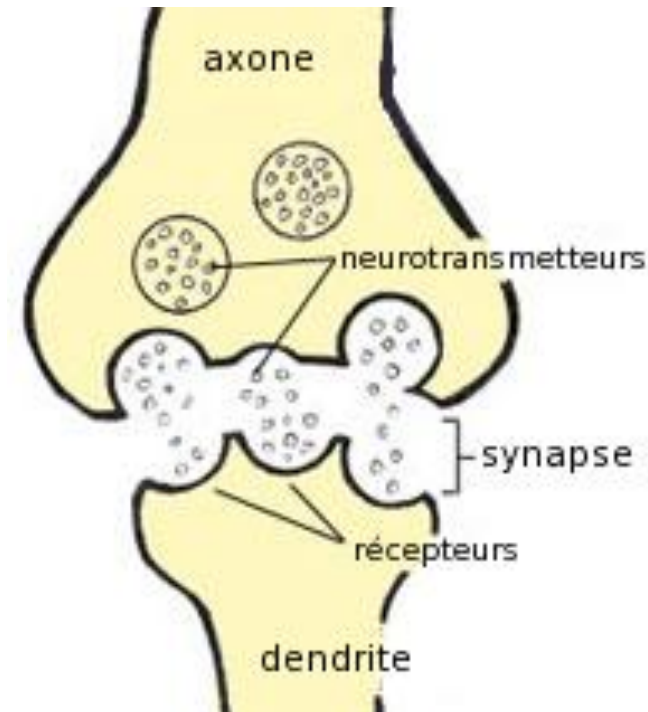
Glutamate [Excitatory]



Glutamate Decarboxylase

γ -aminobutyric acid (GABA) [Inhibitory]

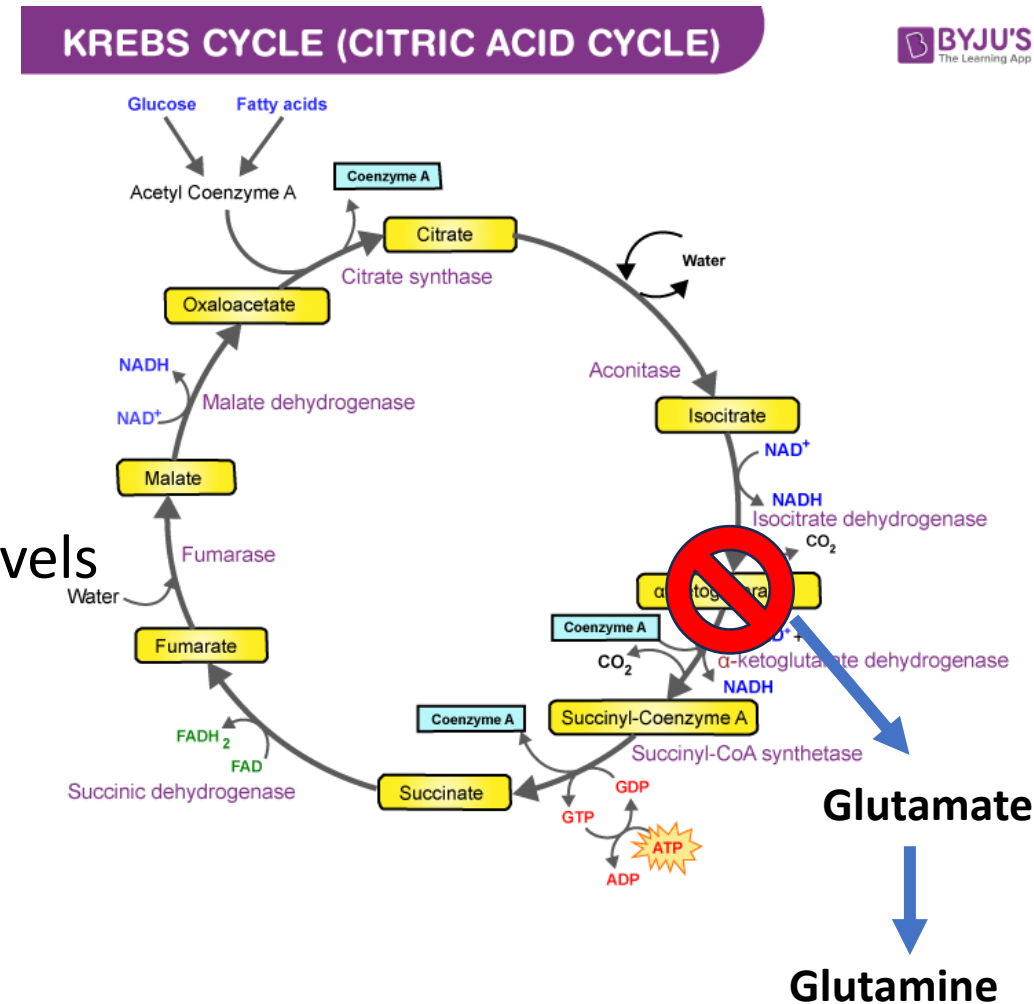
**Main point: Hyperammonemia
alters neurotransmitter balance in
the brain (Excitatory vs Inhibitory)**



Mechanism of ammonia neurotoxicity

Effects of elevated blood ammonia on generation of ATP:

- α -ketoglutarate is converted to glutamate
 - Via glutamate dehydrogenase enzyme
 - Depletion of TCA cycle intermediates
 - **Decreased TCA cycle activity**
 - Reduced ATP synthesis
 - Don't forget, your brain needs ATP
 - Glutamate $\rightarrow$ Glutamine $\rightarrow$ Increased glutamine levels
- Reduced Na⁺/K⁺ ATPase activity
 - Osmotic pressure, edema
 - Alters neurotransmission



- **Some suggested points for students to focus on while studying: Urea cycle**
- Can you describe the transamination (aminotransferase) reaction and its role in the transfer of amino groups. Indicate the importance of vitamin B₆ (pyridoxal phosphate)?
- Can you identify the transport forms of ammonia from peripheral tissues (alanine & glutamine). Indicate the role of glutamate dehydrogenase and glutamine synthetase in glutamine formation?
- Try to describe the role of glutamate dehydrogenase in donating NH₃ for the urea cycle
- Can you list the reactions that form ammonia in the liver?
 - Explain the formation of ammonia in the intestine and its detoxification.
 - Analyze the significance of intestinal ammonia formation in patients with compromised liver function
- Regarding the urea cycle,
 - Identify the sources of nitrogen for the formation of urea
 - Recall the subcellular location of the urea cycle enzymes and energy utilized for the formation of urea
 - Explain the regulation of the urea cycle (regulatory enzyme and mechanisms of regulation) and analyze the significance of N-acetyl glutamate
 - Compare and contrast the five inherited disorders associated with the urea cycle (Specify enzyme deficient, correlate clinical and biochemical features, and generalize the biochemical basis for management of hyperammonemia for each of the disorder)
 - Indicate the biochemical basis for the use of phenylacetate, benzoic acid, arginine and carbamoyl glutamate in the management of the urea cycle disorders
- Regarding acquired hyperammonemia:
 - Identify causes of hyperammonemia
 - Analyze the mechanisms of neurotoxicity of hyperammonemia
 - Explain the biochemical basis of use of antibiotics and lactulose in the management of acquired hyperammonemia